

NTU · Data Analysis and Machine Learning with Python · Spring 2026

WaferDX

An Explainable AI Decision Support System
for Semiconductor Wafer Defect Diagnosis

Abhirup Sain · David Spickenheuer · Martin Werner ·
Matti Lehmann · Staniya Thomas

June 2026

Presentation Outline

Problem & Motivation

- ▶ Semiconductor yield recovery bottleneck
- ▶ Limitations of existing tools
- ▶ What WaferDX solves

System Design

- ▶ Architecture overview
- ▶ 3 key design decisions
- ▶ Processing pipeline

Evaluation

- ▶ 4 model variants · 160 test images
- ▶ Per-class accuracy breakdown
- ▶ Critical failure cases

Insights & Future Work

- ▶ Transferable design principles
- ▶ Governance risks
- ▶ Extensions with more resources

The Problem: Wafer Defect Diagnosis

- ▶ A 300mm fab lot contains **25 wafers**, each passing through **400–600 process steps**
- ▶ Electrical test produces a **binary die map**: pass / fail at every die location
- ▶ The **spatial pattern** of failures — not the count — identifies the responsible tool

Center cluster

→ CVD deposition drift

Linear scratch

→ Robot arm damage

Why speed matters

- ▶ Every undetected faulty step processes **more wafers under failure conditions**
- ▶ Manual map interpretation is **error-prone** and a yield-recovery **bottleneck**

Existing tools either detect anomalies without classifying them, or classify without explaining **which spatial region** drove the decision.

WaferDX: What the System Does



Classify

CNN classifies wafer map against **8 defect types** from WM-811K (811K maps)



Explain

Grad-CAM heatmap shows which spatial region drove the prediction



Advise

Knowledge-based RCA maps defect class → process step → recommended action



Warn

Dual-hypothesis display when confidence gap between top-1 and top-2 < 0.15

Dataset: WM-811K · 811,457 maps · 46,393 lots · 8 defect classes + None

Stack: TensorFlow/Keras (CNN) · Streamlit (UI) · Grad-CAM (explainability) · Python RCA lookup

Input: Upload PNG or draw directly on interactive canvas

System Pipeline

1

Input Upload PNG or draw on canvas



2

Preprocess Grayscale $\rightarrow$ zero-pad $\rightarrow$ nearest-neighbour resize $96 \times 96 \rightarrow$ normalise to $\{0.0, 0.5, 1.0\}$



3

CNN Inference Softmax over 8 defect classes



4

Grad-CAM Gradient of top-class score w.r.t. final conv layer $\rightarrow$ weighted activation heatmap



5

Display Ranked probability bars + heatmap overlay



6

RCA Lookup Defect class → process hypothesis + recommended inspection action



7

Multi-hypothesis If confidence gap < 0.15 → show dual hypothesis panel



8 · Engineer Decides WaferDX never issues a recommendation to act

Seg → CLF Pipeline (rejected as primary)

- ▶ Segmentation network produces per-pixel activation maps per class
 - ▶ Second classifier operates on those maps
 - ▶ **Theoretically** more interpretable — spatial reasoning visible directly
-

Result: 11.9% accuracy

Classification head predicted “**Center**” with **confidence 1.0** for >90% of all inputs — regardless of true class. No discriminative signal from segmentation activations.

Direct CNN (chosen)

- ▶ Single end-to-end optimization problem
 - ▶ Grad-CAM provides mathematically grounded post-hoc spatial attribution
 - ▶ More stable training, simpler debugging
-

Result: 13.1% accuracy (Best CNN)

Root cause of Seg → CLF failure: two components trained **independently**, no end-to-end gradient flow. Joint training would be correct long-term fix.

What happened

Class	No Aug	1. Aug
Edge-Loc	30%	85%
Donut	65%	10%

Edge-Loc: failure can appear at any angular position — orientation invariance **helps**

Donut: radially symmetric ring — rotation removes orientation cues that define the class, **destroying accuracy**

Lesson & correct approach

- ▶ Uniform augmentation is **not a safe default** when class identity is geometrically defined
- ▶ **Class-selective augmentation** is required:

Apply rotation/flip: Edge-Loc, Scratch, Random, Loc
Exclude: Donut, Edge-Ring, Center, Near-full

Not implemented within project timeline — identified post-evaluation. Augmented model retained in UI for class distributions where it helps.

Design Decision 3: Knowledge-Based vs. Learned RCA

Why no learned RCA?

- ▶ Requires labeled causal chains: **wafer map** → **confirmed tool failure**
- ▶ No public dataset links WM-811K labels to confirmed fab process failures
- ▶ SECOM (UCI): sensor data, no spatial maps or defect labels
- ▶ LLM-generated hypotheses: plausible-sounding but **unverifiable** — rejected

Knowledge-based lookup (chosen)

Center	CVD deposition drift; spin-coat blockage
Donut	Non-uniform spin coating; CMP pressure ring
Edge-Ring	RTP thermal non-uniformity
Scratch	Robot end-effector; CMP over-polishing
Near-full	Catastrophic process excursion

Updating for a new failure mode = **edit one dictionary**.
No retraining required.

Scope limit: Mappings calibrated for standard bulk CMOS FEOL. Not applicable to III-V, MEMS, or advanced packaging without revision. All UI outputs labelled as “**pattern hypothesis**”, not “diagnosis”.

Test set

- ▶ 160 PNG images generated from WM-811K via `generate_images.py`
- ▶ **Balanced:** 20 images per class $\times$ 8 classes
- ▶ Single-class only (no multi-defect maps)

Evaluation baseline: random chance on an 8-class balanced set = **12.5%**

4 model variants tested

- ▶ **Best CNN** — direct classifier, no augmentation
- ▶ **Best CNN + Aug** — rotation + flip augmentation
- ▶ **Fullstack** — CNN with full preprocessing pipeline
- ▶ **Seg $\rightarrow$ CLF** — segmentation-then-classify pipeline

`test.py` records: full softmax vector, top-3 labels, correctness flag $\rightarrow$ `evaluation_results.csv`

Evaluation Results: Per-Class Accuracy

Class	Best CNN	1. Aug	Fullstack	Seg→CLF
Center	0%	0%	—	—
Donut	65%	10%	—	—
Edge-Loc	30%	85%	—	—
Edge-Ring	0%	0%	—	—
Loc	0%	0%	—	—
Near-full	5%	15%	—	—
Random	5%	0%	—	—
Scratch	0%	0%	—	—
Overall	13.1%	13.8%	17.5%	11.9%

All models cluster **near random chance (12.5%)** on the balanced test set. High training accuracy does not generalise to rare classes (Scratch: <300 samples in full WM-811K).

Critical Failure Cases

Failure 1: Center / Near-full Confusion

- ▶ Best CNN assigns Near-full confidence **0.90–0.98** to Center inputs
- ▶ **18 out of 20** Center images predicted as Near-full

Operational consequence:

Center → routine CVD calibration drift

Near-full → **catastrophic failure, immediate lot quarantine**

A top-1-only UI would trigger a drastic response for a routine issue **90% of the time.**

High confidence + wrong answer is **more dangerous** than low confidence + wrong answer — the warning threshold would not fire

Failure 2: Augmentation Destroys Donut

- ▶ Adding rotation/flip: Donut drops **65% → 10%**
- ▶ Edge-Loc jumps **30% → 85%** in the same run

Rotation teaches the model a **partial ring is a valid Donut** — conflicting with the class definition.

Failure 3: Seg → CLF Collapse

Pipeline predicts “Center” confidence 1.0 for **>90% of all inputs** regardless of true class — classification head overfit to dominant training class, ignoring segmentation features.

How WaferDX Handles Model Uncertainty



Full Distribution

Always shows **all 8 class probabilities** as a ranked bar chart — not just top-1 label + confidence

Engineer sees: “Donut 57%, Edge-Loc 38%” and can judge calibration before acting



Grad-CAM Heatmap

Overlaid on input image — engineer can verify whether the model is **attending to the actual defect region** or to wafer-edge artifacts

Visual sanity check before initiating any tool inspection



Dual Hypothesis

Auto-triggered when **confidence gap** < 0.15 — displays both process hypotheses side-by-side with confidence scores

“This map is ambiguous between Edge-Ring and Donut — here is what each implies for your process”

All RCA outputs are labelled as “**pattern hypothesis**” throughout the UI — not “diagnosis” or “root cause”. **The engineer decides what to do next.**

Transferable design principles

- ▶ **Augmentation must respect class geometry** — applies to satellite imaging, histopathology, PCB inspection, any spatially structured classification
- ▶ **Calibration > accuracy for decision support** — high confidence + wrong answer is the most dangerous failure mode; temperature/isotonic scaling should be mandatory
- ▶ **Separate classification from domain reasoning** — RCA lookup table updated without model retraining; modularity scales to new failure modes

Governance risks

- ▶ **Calibration failure** — Center/Near-full case shows the confidence-gap warning does not protect against overconfident wrong predictions; per-class calibration on held-out set required
- ▶ **Over-reliance on RCA** — knowledge-based mappings are heuristics, not verified causal chains; cross-check with tool logs mandatory before action
- ▶ **Scope drift** — mappings only valid for standard bulk CMOS FEOL; III-V, MEMS, advanced packaging require different tables

Future Extensions

1 · Near-term

Per-class calibration

500 samples/class calibration set from WM-811K + isotonic regression → fix Center/Near-full overconfidence so threshold warnings fire correctly

2 · Medium-term

Multi-defect maps

Synthetic multi-label maps via logical union of single-class samples → multi-label loss (binary cross-entropy per class) + per-class F1 evaluation

3 · Long-term

Learned RCA

Integrate proprietary fab process logs (tool IDs, chamber sensors, run timestamps) to partially replace heuristic mappings with empirically validated causal associations

Core Contribution

WaferDX shows that explainable decision support requires **more than appending Grad-CAM to a CNN**.

Transparency about model limitations — full distributions, spatial attribution, explicit dual-hypothesis display —
is itself a design feature, not a fallback for low accuracy.

Thank you — questions welcome